Homework 6: Software Testing Fundamentals

CS 150 with Dr. Sam Schwartz

Upload to Canvas when done.

1 Purpose

In this assignment, you will practice the fundamentals of basic software testing in an object-oriented context. You will write unit tests, design testable classes, and understand frameworks for software testing.

2 Tasks

This homework will have a written component (Task A part 1 and part 2) and a coding component (Task B part 1 and part 2). The written component should be submitted as a single pdf file, with code submitted as a .zip file along with screenshots of the output.

Task A – Part 1

This task is part of the written component. Answer the following questions in a PDF document:

1. What is unit testing? What is integration testing? What's the difference between the two?
2. What are test cases, edge cases, and assertions?
3. What, in your opinion, makes for a good test case?

Task B – Part 1

Consider the following code:

```
public class BankAccount {
 private double balance;

 public BankAccount(double initialBalance) {
  this.balance = initialBalance;
 }

 public void deposit(double amount) {
  this.balance += amount;
 }

 public void withdraw(double amount) {
  this.balance -= amount;
 }

 public double getBalance() {
  return this.balance;
 }
}
```

Write unit tests for each method in the above class. Each method should have at least one test associated with an edge case.

Task A – Part 2

The current BankAccount class has problems, given some basic assumptions about how banks work. For example, assumptions like you can't deposit or withdraw negative amounts of money; you can't overdraw your account beyond a certain limit, which is often $0; etc.

Please answer the following question in the PDF document:

4. Please identify problems / violated assumptions which have been found by your unit tests, and how your unit tests caught them.

Task B – Part 2

Update the code provided in Part 1 so that these identified problems are fixed.

Provide screenshots of all your unit tests running and passing.

3 Grading Criteria

In general I am looking for the elements of validity, readability, and fluency in all code-based assignments. (See more below.)

I tend to dock 5ish points off for each error (although smaller or larger quantities like -1pt or -10pts exist based on the magnitude of the error), and will provide free-form feedback detailing why any points were missed in the comments on Canvas.

Please note that I do not get notifications about replies to my comments on Canvas, so if you have any questions please reach out to me directly.

Validity

Student submitted files which implemented the tasks correctly and handled possible errors gracefully.

Moreover, the student submitted screenshot(s) displaying the code's output.

Readability

The deliverable used professional English and typesetting throughout, and had all methods annotated with JavaDoc.

Fluency

The deliverable was executed in such a way that an experienced practitioner would not find the deliverable "weird-in-a-bad-way" or unduly jarring.